

202605 BMDS2133 Image Processing Assignment Specification

This assignment aims to strengthen students' understanding of classical image processing techniques through the development of a practical image analysis solution. Students are required to apply image enhancement, filtering, segmentation, feature extraction, morphology, and object analysis techniques to solve a real-world problem.

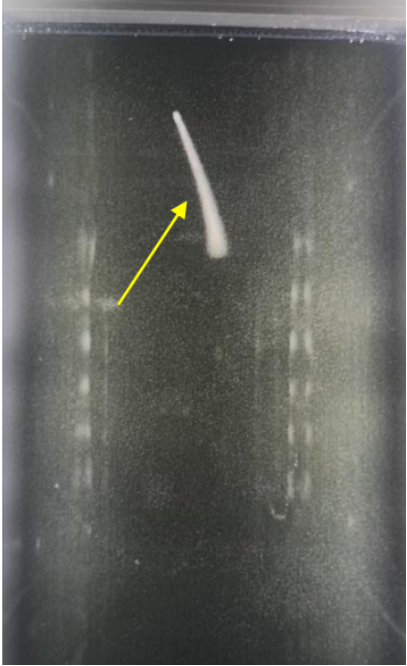
Course Learning Outcomes (CLOs)

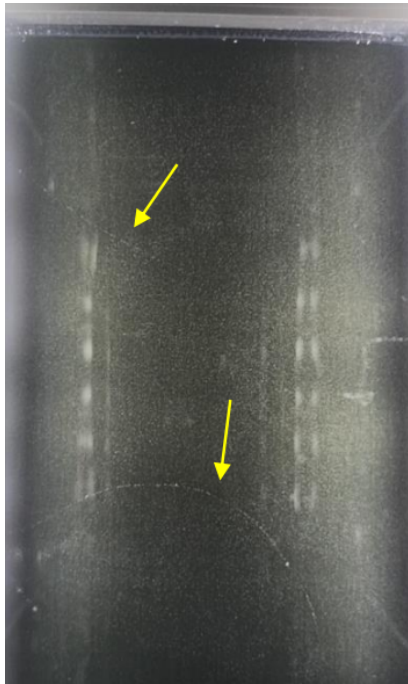
- **CLO2:** Practice suitable image processing techniques and algorithms to solve a given problem. (*P3, PLO3*)
- **CLO3:** Present solution by utilising image processing techniques and tools to address real-world problems. (*A2, PLO10*)

Project Topics and Specific Requirements

The following table outlines the focus and specific scope for each project. Students are expected to define their own technical approaches to achieve these goals, ensuring the system functions as a practical real-world application.

Project Topic	Description	Project-Specific Scope
1. PCB Defect Inspection System	An automated system designed to identify physical manufacturing flaws or missing components on printed circuit boards.	Focus on verifying component placement accuracy, identifying structural inconsistencies, and validating board integrity against a reference model.
2. Fruit Ripeness Assessment	A grading system that classifies fruit maturity levels and quality based on visual surface characteristics.	Focus on surface quality grading, maturity classification, and the quantification of blemishes or damage relative to the total surface area.

3. Fingerprint Enhancement System	A pre-processing pipeline designed to improve the quality of degraded or partial fingerprint images for matching.	Focus on restoring ridge flow patterns, improving image clarity for low-quality inputs, and isolating ridge structures for accurate feature analysis.
4. Cloud Chamber Particle Track Classification	An analysis tool designed to identify, measure, and categorise microscopic particle movement and structural patterns.	Focus on the classification of trajectory types based on geometric metrics, such as track length, width, curvature, and branching behaviour. Alpha Particle Track: Alpha particle tracks are thick and short. It is typically no more than 6 cm long, and their length is quite consistent because it mostly originated from Radon decay. - <i>circumscribed by a Yellow Bounding Box in the dataset given.</i>  Electron Particle Track: Electrons / positrons have the weakest ionisation strength, as a result it has the thinnest track. They are difficult to capture properly but due to their low charge to mass ratio, they bend easily in the magnetic field. In this example, one of

		the electrons is moving in a circle inside my cloud chamber because I put a neodymium magnet below it. The other one on top doesn't seem to bend much because it has higher kinetic energy.through the whole chamber. - <i>circumscribed by a Red Bounding Box in the dataset given.</i>  Dataset: Cloud Chamber Experiment Reference: KEBUK AWAN
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Shared Core Functional Requirements

- **Preprocessing:** Implement noise removal and image enhancement techniques (e.g., Gaussian/Median filtering, contrast stretching).
- **Image Calibration:** Perform spatial scaling and rectification to maintain consistency between pixel dimensions and physical measurements.
- **Object Detection (if applicable):** Perform accurate bounding box and object contour detection.
- **Data Analysis Dashboard:** An interface to summarise object properties and inspection/enhancement results.

- **(Extra Efforts) Reporting:** Automated export of results and findings into PDF format.
- **(Extra Efforts) Video Processing:** Enable ingestion of video streams as input data and execute real-time algorithmic analysis across individual frames.
- **(Extra Efforts) GUI:** Support for bulk image ingestion, such as folder-based uploads or multi-image selection.
- **(Extra Efforts) Supplemental Functionalities:** Integration of advanced features or algorithmic improvements that elevate the overall performance and efficacy of the developed system. You name it.

Assignment Mode

Students may choose ONE of the following assignment modes.

Aspect	Mode A Comparative & Enhancement Study	Mode B Innovative Solution Development
Objective	Investigate, compare, and enhance existing image processing techniques.	Develop an innovative image-processing-based solution for a real-world problem.
Focus	Algorithm comparison and performance improvement.	System development and problem solving.
Problem Selection	Based on one of the provided assignment topics.	May use one of the provided topics or propose a new real-world problem (subject to lecturer approval).
Individual Contribution	Each member studies and implements a different contemporary image processing technique.	Each member develops a different functional module of the system.
Team Contribution	Integrate and compare techniques to produce an improved or hybrid solution.	Integrate all modules into a complete working solution.
Technical Requirement	Implement and benchmark multiple techniques.	Design and develop a complete end-to-end application.

Experimental Evaluation	Compare techniques using quantitative metrics and determine the best-performing approach.	Evaluate effectiveness of the complete solution using suitable performance metrics.
Innovation Requirement	Enhance existing techniques or develop a hybrid approach.	Propose a novel solution, feature, workflow, or application that addresses a practical issue.
Example (Traffic Sign Detection)	Member A: Colour Segmentation Member B: Shape Detection Member C: Hough Transform Member D: Morphological Filtering Compare and combine techniques to improve detection accuracy.	Member A: Image Acquisition Module Member B: Traffic Sign Detection Module Member C: Sign Classification Module Member D: User Interface Module Integrate into a complete driver-assistance prototype.

Requirements

- UK English only.
- Teams are restricted to a maximum of four members. An extension to five members requires strong justification and prior approval from the tutor.
- The following points outline the fundamental expectations for this coursework. Adhering strictly to these guidelines typically results in an Average or Good grade. To secure an Excellent grade (refer to Section *Shared Core Functional Requirements, Extra Efforts*), students require evident extra efforts, such as learning new skills, introducing new ideas, complex Image Processing algorithms, demonstrating the ability to process big data, and/or producing excellent reports with working prototype.
- Proposing a solution that lacks/less fundamental **Image Processing** algorithms will result in the assignment being classified under the poor performance category.
- What to hand in? *Submission of materials is required from the group leader exclusively; all other team members should select "Mark as done" within Google Classroom.*
 - Part 1: Documentation
 - ONE Editable version + ONE PDF
 - Avoid uploading the **originality report**; this document will be produced automatically upon your submission.
 - Part 2: Source Code with presentation slides.
 - Avoid uploading external web-based datasets; instead, provide the citation within the **Part 1 Documentation**. Original data collections are exempt from this requirement.
 - Submit as one ZIP file only

Submission Deadline

Submit the assignment through Google Classroom. Late submission without valid reason will NOT be tolerated. For late submission, there will be a reduction of total marks:

- Late 1 to 3 days after deadline of submission: minus 10 marks
- Late 4 to 7 days after deadline of submission: minus 20 marks
- Late more than 7 days after deadline of submission: 0 marks

In certain circumstances, a student may be allowed to submit the assignment late with valid reason. S/he must contact the tutor at least one week before the assignment is due. The tutor will evaluate whether the circumstance warrants submitting the assignment late, but no guarantee that the students will not be penalised.

Failing to submit the reports & code will lead to failure of the coursework.

Academic Integrity and Plagiarism

Originality remains a critical component of this assignment; consequently, all work must be developed independently, avoiding any unauthorised referencing or copying from other groups. Collaboration is restricted exclusively to your assigned teammates. It is strictly forbidden to share or discuss coursework materials—including your program source code—with individuals outside your team or the course tutor.

Before finalising your submission, ensure total adherence to the **TAR UMT Plagiarism Policy** and the **Free-Rider Policy**. You are also required to include the completed **Group Contract Form** and the **AI usage disclosure form** provided in the documentation template. Any evidence of plagiarism, collusion, or fraudulent actions aimed at obtaining an unmerited academic benefit will lead to severe disciplinary consequences for every student involved.

IMPORTANT: Academic dishonesty will be met with formal disciplinary action.

Red–Yellow–Green Framework

In this course, AI is recognised as a collaborative learning tool. Students are encouraged to utilise AI tools (e.g., ChatGPT, Claude, MidJourney) to support activities such as brainstorming, coding, or refining written work.

● Green (AI-Supported): AI is encouraged for better algorithm structuring, outlining, and tutoring, subject to the requirements of the assigned task.

AI Usage Guidelines: In this course, AI is recognised as a collaborative learning tool. Students are encouraged to utilise AI tools (e.g., ChatGPT, Claude, MidJourney) to support activities such as brainstorming, coding, or refining written work.

Transparency and Accountability: All AI use must be explicitly disclosed. Students are required to include an **AI Disclosure Statement** with each submission, specifying the AI tools and prompts used, as well as the steps taken to verify the accuracy and relevance of the AI-generated content. Students remain fully responsible for the accuracy, logic, and integrity of their final work.

Marking Rubric

Part 1 Documentation

Criteria	Beyond Expectation	Meets Expectation	Approaching Expectation	Missing/Unacceptable
Introduction (10 marks)	Exceptional project background. Objectives are perfectly framed using the SMART framework (Specific, Measurable, Achievable, Relevant, Time-bound). 8 - 10	Clear project background. Objectives are defined and mostly follow the SMART framework. 5 - 7	Vague background context. Objectives are generic, poorly defined, or missing SMART attributes. 2 - 4	Missing background or objectives entirely. 0 - 1
Literature Review (20 marks)	All critical concepts are deeply studied. Includes a clear, comprehensive comparison table contrasting existing works/systems flawlessly. 16 - 20	Core concepts are well studied. Includes a comparison table, though minor details or points of contrast are missing. 11 - 15	Concepts are superficially covered. The comparison table is messy, incomplete, or missing. 6 - 10	No literature review or comparison table provided. 0 - 5
Methodology (20 marks)	Includes a detailed, logical flowchart paired with an exceptionally thorough, step-by-step explanation of the implementation process. 16 - 20	Includes an understandable flowchart with a clear explanation of the procedural steps. 11 - 15	Flowchart is confusing, inaccurate, or poorly explained in the accompanying text. 6 - 10	Missing methodology, flowchart, or description of the process. 0 - 5

Result & Discussion (20 marks)	Displays robust experimental results alongside a detailed, critical comparison and analysis of the data outcomes. 16 - 20	Displays clear experimental results with a logical discussion and analysis of the findings. 11 - 15	Results are presented but lack meaningful experimental data, detailed comparison, or depth in analysis. 6 - 10	Experimental results and analysis are completely missing or fabricated. 0 - 5
Conclusion (10 marks)	Masterfully synthesises findings and explicitly links back to every SMART objective stated in the introduction. 8 - 10	Clearly summarises main findings and links them back to the project objectives. 5 - 7	Weak conclusion that fails to clearly connect the final outcomes back to the original objectives. 2 - 4	Missing conclusion or fails to address the project outcomes. 0 - 1
Spelling, Grammar & Writing Mechanics (10 marks)	Written flawlessly in UK English (e.g., <i>analyse, colour, programme</i>). Zero use of first-person pronouns (<i>I, we, my, our</i>); uses strict objective/passive voice. 8 - 10	Written in UK English with rare errors. Strictly avoids first-person pronouns with only 1-2 accidental slips. 5 - 7	Inconsistent spelling styles (mixes US/UK). Frequent grammatical errors or improper use of first-person pronouns. 2 - 4	Poorly written, fails UK English standards, or heavily written in the first person. 0 - 1
References (10 marks)	Flawless APA reference style throughout. Zero incomplete citations; every in-text citation matches the reference list perfectly. 8 - 10	Consistently formatted in APA style with minor punctuation errors. All citations are complete. 5 - 7	Inconsistent APA formatting, missing publication details, or incomplete/broken citations. 2 - 4	No references provided, or widespread plagiarism is evident. 0 - 1

Part 2 Prototype

Criteria	Good	Accomplished	Poor
User Interface / Output (20%)	The UI is highly intuitive, professional, and visually appealing. Output is accurate, dynamic, and error-free. 15 - 20	The UI is functional and user-friendly. Output is mostly correct with minor aesthetic or minor display bugs. 9 - 14	The UI is poorly designed, confusing, or broken. Output is incorrect, messy, or missing. 0 - 8
Programming (30%)	Code is exceptionally clean, well-commented, modular, and optimised. Follow industry best practices. 21 - 30	Code is functional and organised, but contains some redundancies or lacks sufficient comments. 11 - 20	Code is disorganised, inefficient, hard to read, or heavily reliant on hardcoding. 0 - 10
Degree of Completion (20%)	All planned features and requirements are fully implemented and working perfectly. 15 - 20	Core features are fully functional, but some secondary features are incomplete or missing. 9 - 14	Prototype is largely incomplete; major core functionalities do not work. 0 - 8
System Implementation (20%)	Seamless deployment and integration of system components (databases, APIs, etc.). Highly stable. 15 - 20	The system is successfully implemented and integrated, but experiences occasional stability issues. 9 - 14	Failed implementation. Components do not integrate, or the system crashes frequently. 0 - 8
Understanding of Code / On-the-spot Coding (10%)	The student demonstrates mastery of the code, answers all technical questions confidently, and completes live tasks effortlessly. 8 - 10	The student understands the core logic and answers most questions, but struggles slightly with complex on-the-spot tasks. 4 - 7	The student cannot explain their own code or fail to perform basic live coding modifications. 0 - 3